A RING DERIVED COMPLETE WITH RESPECT TO AN IDEAL FORMS A HENSELIAN PAIR

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Allowed standard prerequisites (editorial).

- A pair is henselian as soon as it is henselian with respect to every principal ideal generated by an element of the ideal. invoked at proof lines 27092-27094. Reduces the statement for the given ideal to the case of a single element.
- A complete ring with respect to a principal ideal forms a henselian pair, and a pair is henselian when its completion is and the kernel of the completion map is handled. invoked at proof lines 27096-27102. Transfers the property from the adic completion back to the ring through the two intermediate statements.
- A ring that is derived complete with respect to an element is derived complete, hence complete, with respect to the principal ideal that element generates. used at proof lines 27094-27096. Justifies applying the completeness statements to the principal ideal generated by the chosen element.

Source reference. Original lemma 0G1S: sources/stacks-more-algebra/pinned-original-source.tex, source lines 2938–2938. Invoked at proof lines 27092-27094 to reduce the statement to the ideal generated by a single element; every step after the reduction is inside the counted span.

Source reference. Original definition 091S: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26021–26021. Cited at proof lines 27094-27096 for the immediate consequence that the ring is complete with respect to the principal ideal; the use made of that completeness is inside the counted span.

Source reference. Original lemma 091R: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25963–25963. Used at proof lines 27096-27098 to see that the map from the ring to its completion is surjective; the transfer of henselianity that follows is inside the counted span.

Source reference. Original lemma 0ALJ: sources/stacks-more-algebra/pinned-original-source.tex, source lines 2483–2483. Invoked at proof lines 27098-27100 to see that the pair with the completion is henselian; the reduction to the intersection of the powers that uses it is inside the counted span.

Source reference. Original lemma 0DYD: sources/stacks-more-algebra/pinned-original-source.tex, source lines 2766–2766. Cited at proof lines 27100-27102 to reduce to the

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pair formed with the intersection of the powers of the ideal; the final two references that settle that case are inside the counted span.

Source reference. Original lemma 0G3G: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27068–27068. One of the two statements invoked at proof lines 27102-27103 to finish; the reduction stage that makes them applicable is inside the counted span.

Source reference. Original lemma 0ALI: sources/stacks-more-algebra/pinned-original-source.tex, source lines 2417–2417. The second of the two statements invoked at proof lines 27102-27103 to finish the proof; the preceding reductions are inside the counted span.

Lemma 0.2. *Let A be a ring derived complete with respect to an ideal I. Then (A, I) is a henselian pair.*

Definition 0.3. A *henselian pair* is a pair (A, I) satisfying

(1) I is contained in the Jacobson radical of A, and
(2) for any monic polynomial $f \in A[T]$ and factorization $\overline{f} = g_0 h_0$ with $g_0, h_0 \in A/I[T]$ monic generating the unit ideal in $A/I[T]$, there exists a factorization $f = gh$ in $A[T]$ with g, h monic and $g_0 = \overline{g}$ and $h_0 = \overline{h}$.

Proof. Let $f \in I$. By Lemma 0G1S it suffices to show that (A, fA) is a henselian pair. Observe that A is derived complete with respect to fA (follows immediately from Definition 091S). By Lemma 091R the map from A to the f-adic completion A' of A is surjective. By Lemma 0ALJ the pair (A', fA') is henselian. Thus it suffices to show that $(A, \bigcap f^n A)$ is a henselian pair, see Lemma 0DYD. This follows from Lemmas 0G3G and 0ALI. □